

State-action value function example

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1 State Action Value Function Example

In this Jupyter notebook, you can modify the mars rover example to see how the values of $Q(s,a)$ will change depending on the rewards and discount factor changing.

```
[1]: import numpy as np
      from utils import *
```

```
[2]: # Do not modify
      num_states = 6
      num_actions = 2
```

```
[3]: terminal_left_reward = 100
      terminal_right_reward = 40
      each_step_reward = 0

      # Discount factor
      gamma = 0.5

      # Probability of going in the wrong direction
      misstep_prob = 0
```

```
[4]: generate_visualization(terminal_left_reward, terminal_right_reward,
      ↪each_step_reward, gamma, misstep_prob)
```

Optimal policy					
100.0	50.0	25.0	12.5	20.0	40.0
	←	←	←	→	
100	0	0	0	0	40

Q(s,a)					
100.0 100.0	50.0 12.5	25.0 6.25	12.5 10.0	6.25 20.0	40.0 40.0
100	0	0	0	0	40

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